

2.1.1. Roots of a Quadratic Equation

ALGORITHM

Step 1 : Start.

Step 2 : Read the coefficients a, b, and c.

Step 3 : Calculate the discriminant using:- $D=b^2-4ac$.

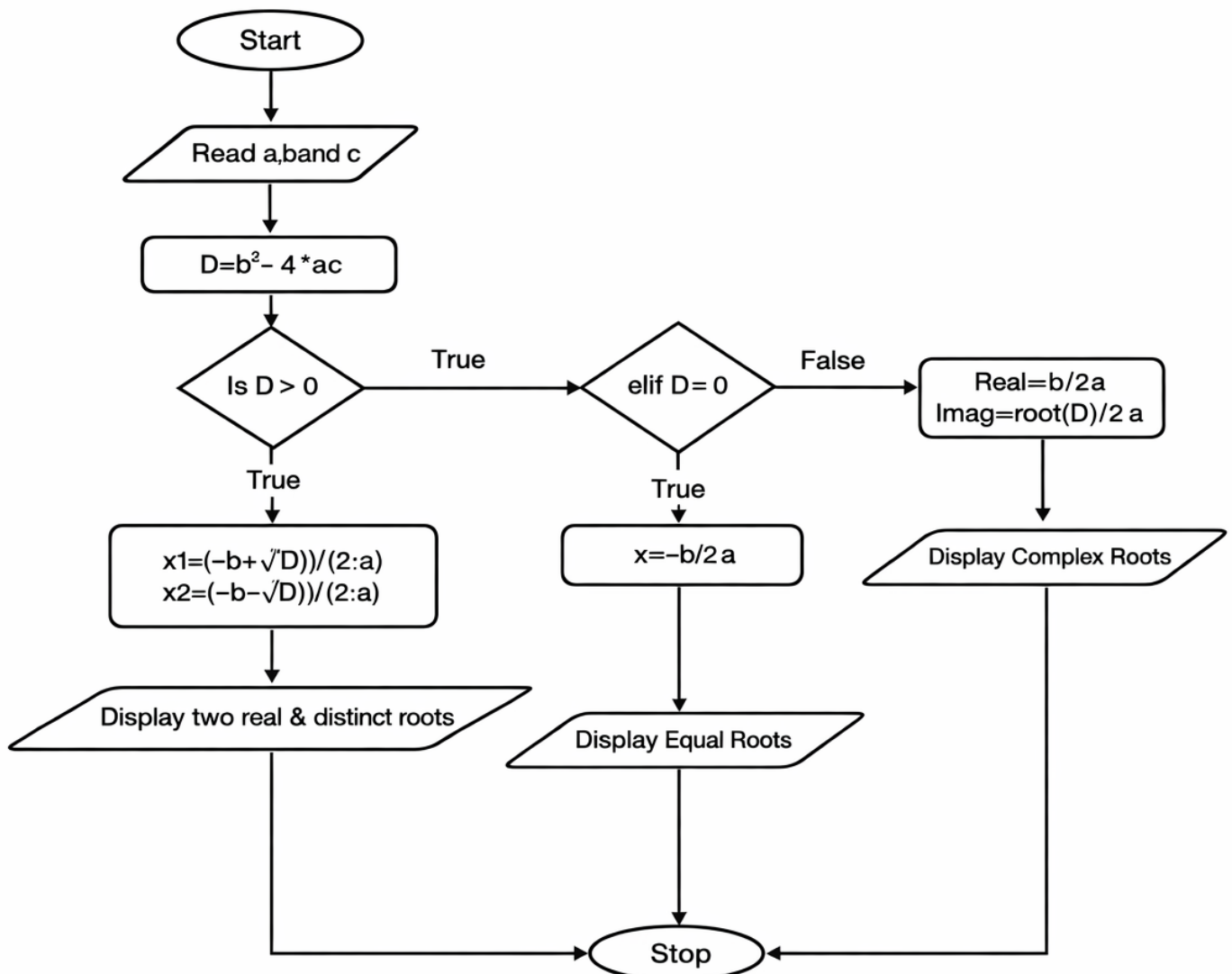
Step 4 : If $D > 0$ Compute:- $x1=(-b+\sqrt{D})/2a$ and $x2=(-b-\sqrt{D})/2a$ Display two real and distinct roots.

Step 5 : Else if $D = 0$ Compute:- $x=-b/2a$ Display equal real roots.

Step 6 : Else ($D < 0$) Compute:- Real part = $-b/2a$ Imaginary part = $\sqrt{-D}/2a$ Step 1 : Display complex roots.

Step 7 : Stop.

FLOWCHART



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2.1.1. Roots of a Quadratic Equation00:25

Write a program to find the roots of a quadratic equation, given its coefficients a , b , and c . Use the quadratic formula: $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a , b , and c , respectively.

Output Format:

- If roots are real and different, print:
root1 = <Root1>
root2 = <Root2>

If roots are the same, print:

Sample Test Cases

quadratic...

```
1 a,b,c = map(int, input().split())
2
3 d = (b * b) - (4 * a * c)
4
5 if d > 0:
6     sq = d ** 0.5
7     root1 = (-b + sq) / (2 * a)
8     root2 = (-b - sq) / (2 * a)
9     print(f"root1 = {root1:.2f}")
10    print(f"root2 = {root2:.2f}")
```

Average time0.002 s2.17 ms

Maximum time0.004 s4.00 ms

3 out of 3 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 14 ms

Debug

Expected output	Actual output
1 -5 6	1 -5 6
root1 = 3.00	root1 = 3.00
root2 = 2.00	root2 = 2.00

Test case 22 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >